

```
import pandas as pd
```

1. Create a DataFrame by reading in the 2019_Yellow_Taxi_Trip_Data.csv file. Examine the first 5 rows.

```
url = "https://data.cityofnewyork.us/resource/2upf-qytp.csv"
df = pd.read_csv(url)
df.columns = df.columns.str.lower().str.replace(' ', '_')
print(df.head())
```

	vendorid	tpep_pickup_datetime	tpep_dropoff_datetime	
0	1	2019-01-01T00:46:40.000	2019-01-01T00:53:20.000	
1	1	2019-01-01T00:59:47.000	2019-01-01T01:18:59.000	
2	2	2018-12-21T13:48:30.000	2018-12-21T13:52:40.000	
3	2	2018-11-28T15:52:25.000	2018-11-28T15:55:45.000	
4	2	2018-11-28T15:56:57.000	2018-11-28T15:58:33.000	

	passenger_count	trip_distance	ratecodeid	store_and_fwd_flag	
0	1	1.5	1		N
1	1	2.6	1		N
2	3	0.0	1		N
3	5	0.0	1		N
4	5	0.0	2		N

	pulocationid	dolocationid	payment_type	fare_amount	extra
0	151	239	1	7.0	0.5
1	239	246	1	14.0	0.5
2	236	236	1	4.5	0.5
3	193	193	2	3.5	0.5
4	193	193	2	52.0	0.0

	tip_amount	tolls_amount	improvement_surcharge	total_amount	
0	1.65	0.0	0.3	9.95	
1	1.00	0.0	0.3	16.30	
2	0.00	0.0	0.3	5.80	
3	0.00	0.0	0.3	7.55	
4	0.00	0.0	0.3	55.55	

	congestion_surcharge
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN

1. Find the dimensions (number of rows and number of columns) in the data.

```
rows, cols = df.shape
print(f"DIMENSIONS \n Rows: {rows}, Columns: {cols}")
```

DIMENSIONS

Rows: 1000, Columns: 18

1. Using the data in the 2019_Yellow_Taxi_Trip_Data.csv file, calculate summary statistics for the fare_amount, tip_amount, tolls_amount, and total_amount columns.

```
print(df[['fare_amount', 'tip_amount', 'tolls_amount',
'total_amount']].describe())
```

	fare_amount	tip_amount	tolls_amount	total_amount
count	1000.000000	1000.000000	1000.000000	1000.000000
mean	12.540500	1.705250	0.144560	15.703410
std	9.525248	2.178548	0.899814	10.897677
min	-2.500000	0.000000	0.000000	-3.800000
25%	6.500000	0.000000	0.000000	8.800000
50%	9.500000	1.260000	0.000000	12.360000
75%	15.500000	2.460000	0.000000	19.125000
max	82.500000	25.000000	5.760000	83.800000

1. Isolate the fare_amount, tip_amount, tolls_amount, and total_amount for the longest trip by distance (trip_distance).

```
idx = df['trip_distance'].idxmax() # find max
longest_trip = df.loc[idx, ['fare_amount', 'tip_amount',
'tolls_amount', 'total_amount']] # using pandas df.loc
print("\nLongest trip values:")
print(longest_trip)
print("Longest trip distance:", df.loc[idx, 'trip_distance'])
```

```
Longest trip values:
fare_amount      82.5
tip_amount        0.0
tolls_amount      0.0
total_amount     83.8
Name: 319, dtype: object
Longest trip distance: 31.57
```